

Maxwell's eqns

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$

Note: • the del vector (operator)

$$\vec{\nabla} = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$$

• Gradient of a function f : $\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}$

• Divergence of a vector $\vec{A}$: $\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} \hat{x} + \frac{\partial A_y}{\partial y} \hat{y} + \frac{\partial A_z}{\partial z} \hat{z}$

• Curl of a vector $\vec{A}$:

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} - \left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z}$$

• Just useful to remember $\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - (\vec{\nabla} \cdot \vec{\nabla}) \vec{A}$

the Laplacian $\nabla^2 \vec{A} = \frac{\partial^2 \vec{A}}{\partial x^2} + \frac{\partial^2 \vec{A}}{\partial y^2} + \frac{\partial^2 \vec{A}}{\partial z^2}$

Source: Youtube Lecture by Prof. Greg Durgin (WA-Vol: Maxwell's Eqs)

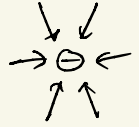
① Maxwell's Eqs - Integral Form

Electric charges spawn electric flux



positive charges spawn

negative charges sink



Gauss's Law

We also know Ampere's Law

$$\oint_L \vec{H} \cdot d\vec{\ell} = \oint_S \vec{J} \cdot d\vec{a}$$

$$\vec{\nabla} \times \vec{H} = \vec{J}$$

Maxwell's Four Equations: (Fully describe Electromagnetism)

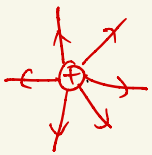
$$\textcircled{1} \oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0} = \int_V \frac{\rho dV}{\epsilon_0}$$

Gauss law for Electricity

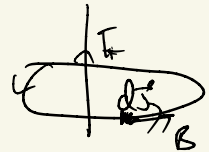
$$\textcircled{2} \oint \vec{B} \cdot d\vec{A} = 0$$

Gauss law for Magnetism

There are NO magnetic monopoles

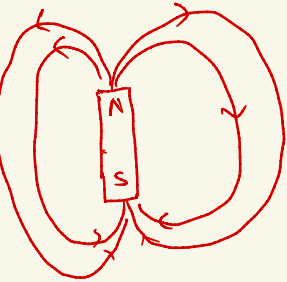


$$E(r) \propto \frac{1}{r^2}$$



$$B \cdot 2\pi r = \mu_0 I$$

$$B(r) = \frac{\mu_0 I}{2\pi r}$$



only magnetic dipoles, no monopoles

$\textcircled{3}$ Закон Фарадея

$$\boxed{EMF = -\frac{d\Phi_B}{dt}}$$

$$\oint \vec{E} \cdot d\vec{x} = -\frac{d\Phi_B}{dt}$$

line integral



$$\vec{E} \cdot d\vec{s} = |\vec{E}| dl \cos\theta$$

$\textcircled{4}$ Закон Ампера (с поправкой Максвелла)

$$\star \oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc} + \mu_0 \left(\epsilon \frac{d\Phi_E}{dt} \right)$$

displacement current
 (в отсутствие б конвекционных)

Electric Fields

$$EMF = - \frac{d\Phi_B}{dt}$$

- charges

$$\frac{d\Phi_E}{dt}$$

Magnetic Fields

- Currents

$$\frac{d\Phi_E}{dt}$$

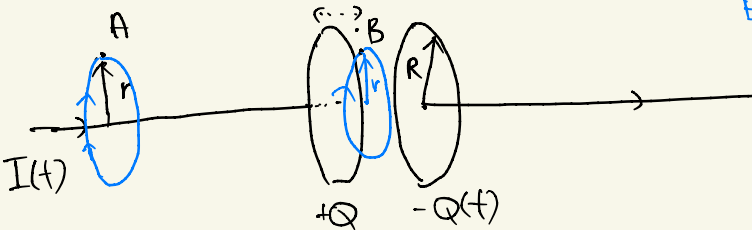


Ex: Displacement Current

$$Q = CV$$

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0}$$

$$E(t) = \frac{Q(t)}{A\epsilon_0}$$



At point A:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc}$$

$$\vec{B}(r) 2\pi r = \mu_0 I(t)$$

$$B(r) = \frac{\mu_0 I(t)}{2\pi r}$$

At Point B $\rightarrow \infty$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc}$$

$$B = 0?$$

Idisp. current

$$B 2\pi r = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$= \mu_0 \epsilon_0 \frac{d}{dt} \left(\frac{Q}{A\epsilon_0} \right) = \mu_0 I$$

Other important things that you will learn in

future:

• Boundary Conditions *

• Material Properties

↳ conductor

↳ insulator (i.e., dielectric)

Quantum
Mechanics

• Spins

↳ Ferromagnetism

↳ Paramagnetism

↳ Diamagnetism

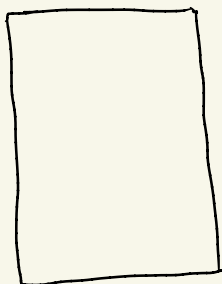
↳ Hysteresis

(Modern Physics)

↑
↑ paper, Einstein etc

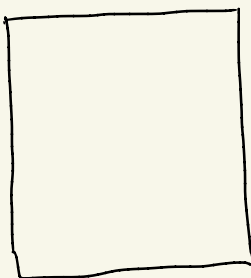
Maxwell's Eqns

Word Form



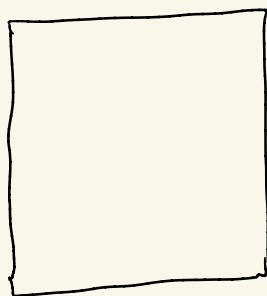
✓
Resnick

Integral Form

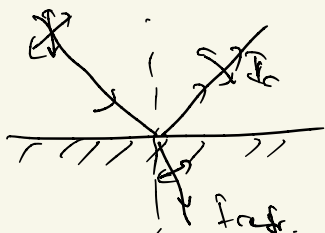


✓
Resnick

Differential Form



↑
Purcell & Griffiths
Griffiths



Feynman eqns

10PH ISPHO 2024

→ Maxwell's eqns

Name	Integral equations	Differential equations	Meaning
Gauss's law	$\oiint_{\partial\Omega} \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\epsilon_0} \iiint_{\Omega} \rho dV$	$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$	The electric field leaving a volume is proportional to the charge inside.
Gauss's law for magnetism	$\oiint_{\partial\Omega} \mathbf{B} \cdot d\mathbf{S} = 0$	$\nabla \cdot \mathbf{B} = 0$	There are no magnetic monopoles; the total magnetic flux piercing a closed surface is zero.
Maxwell–Faraday equation (Faraday's law of induction)	$\oint_{\partial\Sigma} \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \iint_{\Sigma} \mathbf{B} \cdot d\mathbf{S}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	The voltage accumulated around a closed circuit is proportional to the time rate of change of the magnetic flux it encloses.
Ampère's circuital law (with Maxwell's addition)	$\oint_{\partial\Sigma} \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 \iint_{\Sigma} \mathbf{J} \cdot d\mathbf{S} + \mu_0 \epsilon_0 \frac{d}{dt} \iint_{\Sigma} \mathbf{E} \cdot d\mathbf{S}$	$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$	Electric currents and changes in electric fields are proportional to the magnetic field circulating about the area they pierce.

Electric Fields

$$\mathcal{E}MF = - \frac{d\Phi_B}{dt}$$

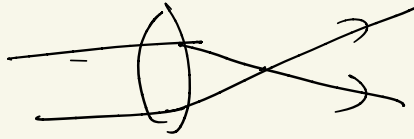
- charges
- $\frac{d\Phi_E}{dt}$

Magnetic Fields

- Currents
- $\frac{d\Phi_E}{dt}$ ★

Оптика

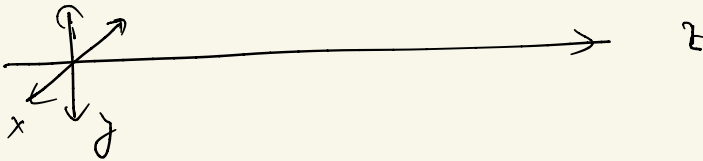
Геометрическая
10%



Волновая 90%

исходит из уравнений
Максвелла

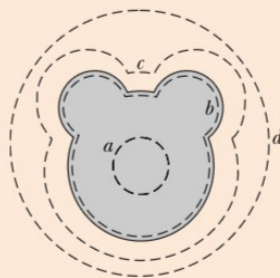
$$\vec{E}(r,t) = (E_x \hat{i} + E_y \hat{j}) e^{-i(kz - \omega t)}$$



Class Resume at 3:02 PM

CHECKPOINT 3

The figure is a view of one plate of a parallel-plate capacitor from within the capacitor. The dashed lines show four integration paths (path b follows the edge of the plate). Rank the paths according to the magnitude of $\oint \vec{B} \cdot d\vec{s}$ along the paths during the discharging of the capacitor, greatest first.



$$\boxed{\oint \vec{B} \cdot d\vec{s}} = \underbrace{\mu_0 I_{enc.}}$$

line integral

Ampere - Maxwell Law:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \left(I_{enc.} + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

$$\boxed{a < b, c, d}$$

$$\boxed{b, c, d > a}$$

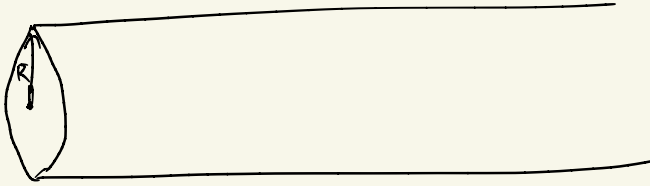
Table 32-1

Maxwell's Equations^a

Name	Equation	
Gauss' law for electricity	$\oint \vec{E} \cdot d\vec{A} = q_{enc}/\epsilon_0$	Relates net electric flux to net enclosed electric charge
Gauss' law for magnetism	$\oint \vec{B} \cdot d\vec{A} = 0$	Relates net magnetic flux to net enclosed magnetic charge
Faraday's law	$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$	Relates induced electric field to changing magnetic flux
Ampere - Maxwell law	$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} + \mu_0 i_{enc}$	Relates induced magnetic field to changing electric flux and to current

^aWritten on the assumption that no dielectric or magnetic materials are present.

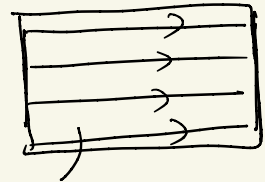
•17 A silver wire has resistivity $\rho = 1.62 \times 10^{-8} \Omega \cdot \text{m}$ and a cross-sectional area of 5.00 mm^2 . The current in the wire is uniform and changing at the rate of 2000 A/s when the current is 100 A . (a) What is the magnitude of the (uniform) electric field in the wire when the current in the wire is 100 A ? (b) What is the displacement current in the wire at that time? (c) What is the ratio of the magnitude of the magnetic field due to the displacement current to that due to the current at a distance r from the wire?



$$\rho = 1.62 \times 10^{-8} \Omega \cdot \text{m}$$

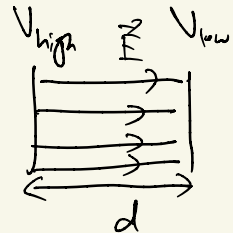
$$A = 5.00 \text{ mm}^2$$

$$\frac{dI}{dt} = 2000 \frac{\text{A}}{\text{s}}, \quad I_0 = 100 \text{ A}$$



$$\sigma = \frac{1}{\rho}$$

(a) $E = ?$



Ohm's law

★

$$\vec{J} = \sigma \vec{E}$$

$$\frac{I}{\text{Area}} = \frac{E}{\rho} \rightarrow$$

$$E = \rho \frac{I}{\text{Area}}$$

(17b) $\oint \vec{B} \cdot d\vec{s} = \mu_0 \left(I_{enc} + \underbrace{\epsilon_0 \frac{d\Phi_E}{dt}}_{I_{disp.}} \right)$

$I_{disp.} = \epsilon_0 \frac{d}{dt} (\Phi_E)$ ✓

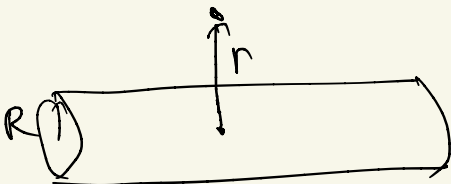
$\Phi_E = EA$

$= \left(\frac{\rho I}{A} \right) A$

$= \rho I$

$I_{disp.} = \epsilon_0 \frac{d}{dt} (\rho I) = \rho \epsilon_0 \frac{dI}{dt}$ ✓

(17c) $\frac{B_{enc.}}{B_{I_{disp.}}} = \frac{\frac{(\cancel{\mu_0} I_0)}{2\pi r}}{\frac{\cancel{\mu_0} I_{disp.}}{2\pi r}} = \frac{I_0}{\rho \epsilon_0 \frac{dI}{dt}}$ ✓



•18 The circuit in Fig. 32-30 consists of switch S, a 12.0 V ideal battery, a 20.0 M Ω resistor, and an air-filled capacitor. The capacitor has parallel circular plates of radius 5.00 cm, separated by 3.00 mm. At time $t = 0$, switch S is closed to begin charging the capacitor. The electric field between the plates is uniform. At $t = 250 \mu\text{s}$, what is the magnitude of the magnetic field within the capacitor, at radial distance 3.00 cm?

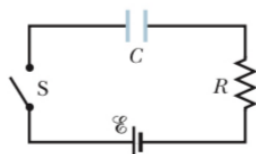
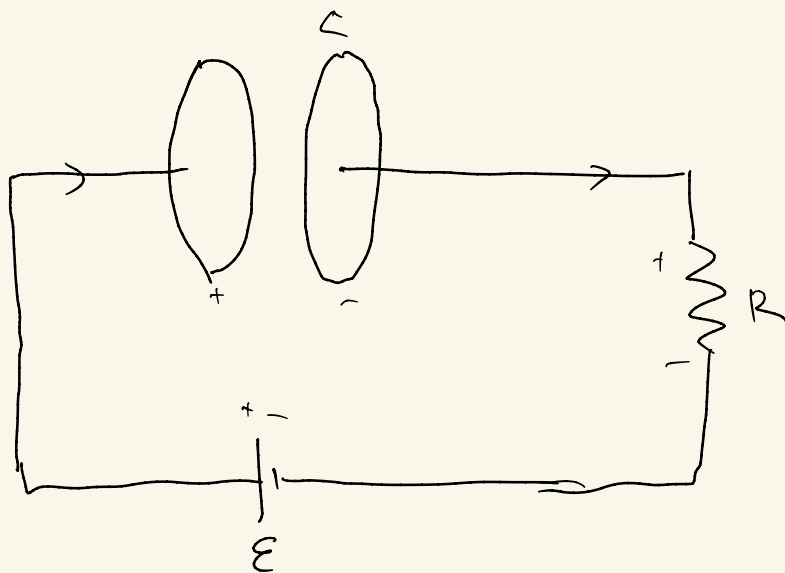


Fig. 32-30 Problem 18.



$$\mathcal{E} - \frac{Q}{C} - IR = 0$$

$$\mathcal{E} - \frac{Q}{C} - \frac{dQ}{dt} R = 0$$

$$\frac{\mathcal{E}}{R} - \frac{Q}{RC} = \frac{dQ}{dt}$$

$$dt = \left(\frac{RC}{\mathcal{E}C - Q} \right) dQ \rightarrow$$

$$-\frac{dt}{RC} = \frac{dQ}{Q - C\mathcal{E}}$$

$$-\int_0^t \frac{dt}{RC} = \int_0^Q \frac{dQ}{Q - CE}$$

Method of Substitution

$$-\frac{t}{RC} = \int_{u_i}^{u_f} \frac{du}{u}$$

$$\text{let } u = Q - CE$$

$$\frac{du}{dQ} = 1 - 0$$

$$-\frac{t}{RC} = \ln u = \ln(Q - CE) \Big|_0^Q \quad dQ = du$$

$$-\frac{t}{RC} = \ln(Q - CE) - \ln(-CE)$$

$\uparrow$
undefined $\times$

$$-\int_0^t \frac{dt}{RC} = -\int_0^Q \frac{dQ}{(CE - Q)}$$

so use $u = CE - Q$, not $Q - CE$

$$-\frac{t}{RC} = \int_{u_i}^{u_f} \frac{du}{u}$$

$$u = CE - Q$$

$$\frac{du}{dQ} = -1$$

$$-\frac{t}{RC} = \ln(CE - Q) \Big|_0^{Q(t)} = \underbrace{\ln[CE - Q(t)] - \ln[CE]}$$

$$-\frac{t}{RC} = \ln \left[\frac{CE - Q(t)}{CE} \right]$$

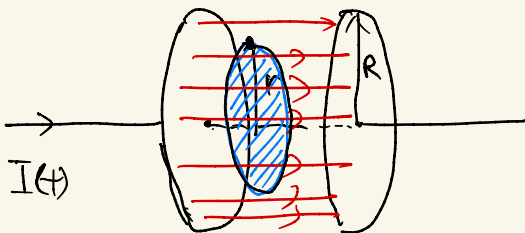
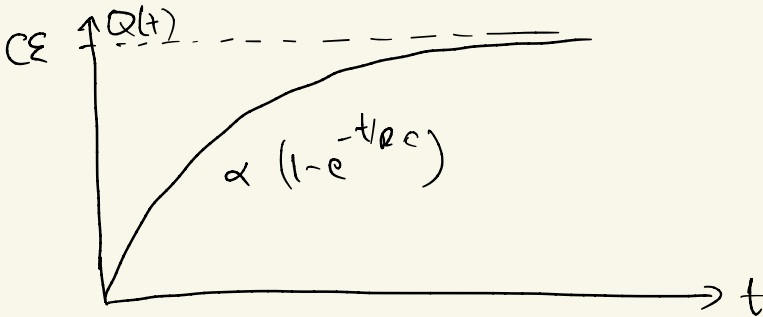
$$-\frac{t}{RC} = \ln \left[\frac{CE - Q(t)}{CE} \right]$$

To remove logarithm, apply exponential to both sides:

$$e^{-t/RC} = \frac{CE - Q(t)}{CE}$$

$$CE e^{-t/RC} = CE - Q(t)$$

$$Q(t) = CE (1 - e^{-t/RC})$$



$$\oint \vec{B}(r) \cdot d\vec{s} = \mu_0 \left(I_{enc} + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

$$B(r) (2\pi r) = \mu_0 \left(0 + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

3 gr 6 $A = \pi R^2$, normally $\sigma = \frac{Q}{A} = \frac{Q}{\pi R^2}$

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{A \epsilon_0}$$

★

$$B(r) [2\pi r] = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\Phi_E = EA = E \pi r^2$$

we can do this without integration because current & E-field is uniformly distributed

$$= \left(\frac{Q}{A \epsilon_0} \right) \pi r^2$$

$$= \frac{Q}{(\pi R^2 \epsilon_0)} \pi r^2$$

$$= \frac{Q r^2}{\epsilon_0 R^2}$$

$$B(r) [2\pi r] = \mu_0 \cancel{\epsilon_0} \frac{dQ}{dt} \left(\frac{r^2}{\cancel{\epsilon_0} R^2} \right)$$

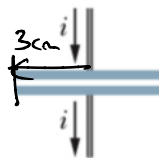
Plug back $Q(t) = C\mathcal{E}(1 - e^{-t/RC})$.

$$\frac{dQ}{dt} = \frac{d}{dt} (C\mathcal{E} - C\mathcal{E} e^{-t/RC}) = 0 - \frac{C\mathcal{E}}{(-RC)} e^{-t/RC}$$

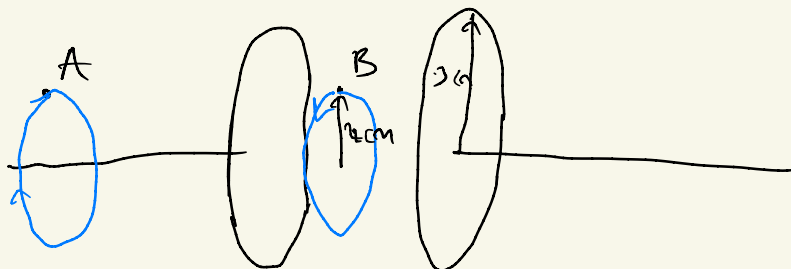
$$= \frac{\mathcal{E}}{R} e^{-t/RC}$$

$$B(r) = \frac{1}{2\pi r} \underbrace{\left(\frac{\mu_0 r^2}{R^2} \right)}_{\text{capacitor's radius}} \underbrace{\left(\frac{\mathcal{E}}{R} e^{-t/RC} \right)}_{\text{resistance}}$$

••22  **Nonuniform displacement current.** Figure 32-29 shows a circular region of radius $R = 3.00$ cm in which a displacement current i_d is directed out of the page. The magnitude of the displacement current is given by $i_d = (3.00 \text{ A})(r/R)$, where r is the radial distance ($r \leq R$). What is the magnitude of the magnetic field due to i_d at radial distances (a) 2.00 cm and (b) 5.00 cm?



••23 **SSM ILW** In Fig. 32-31, a parallel-plate



$$B 2\pi r = \mu_0 I_{enc}$$

$$B(r) = \frac{\mu_0 I_{enc}}{2\pi r}$$

At point A.

At point B
according to Ampere's Law
you should get $B = 0$

BUT measurements suggest
 $B \neq 0$ at B!

Maxwell then explained this
with

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

And he named $\epsilon_0 \frac{d\Phi_E}{dt}$ term as a
displacement current, which is not a real one.
It is just a changing electric flux!

22 solution

Start from Maxwell-Ampere law:

$$\oint \vec{B}(r) \cdot d\vec{s} = \mu_0 (\cancel{I_{enc}} + I_{disp.}) \quad \rightarrow \quad \epsilon_0 \frac{d\Phi_E}{dt}$$

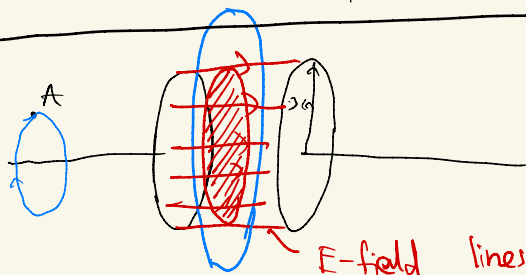
We're given
$$i_{disp.} = 3A \left(\frac{r}{R} \right)$$

$$B 2\pi r = \mu_0 i_{disp.}(r)$$

$$B(r) = \frac{\mu_0 i_{disp.}(r)}{2\pi r}$$

a)
$$B(r=2\text{cm}) = \frac{\mu_0}{2\pi r} 3A \left(\frac{2\text{cm}}{3\text{cm}} \right) = \frac{\mu_0 2A}{2\pi r}$$

b)
$$B(r=5\text{cm}) = \frac{\mu_0}{2\pi r} (3A) = \frac{\mu_0 (3A)}{2\pi r} \quad \text{when } (r > R)$$



HW ch 32

Problems: 3, 4, 8, 9, 10, 11, 17, 18, 21, 25, 27, 29

Deadline: Sep 18, 2025

<https://youtu.be/6WyFcA3m0Es?feature=shared>

Handwritten physics notes on a blackboard background, showing the derivation of the magnetic field B for a parallel plate capacitor with a time-varying electric field.

Diagram: A parallel plate capacitor with plates of area A and separation d . The electric field E is shown between the plates. The magnetic field B is shown as concentric circles around the axis of the capacitor.

Equations and derivations:

- $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left[i_{\text{enc}} + \epsilon_0 \frac{d\Phi_E}{dt} \right]$ (Ampere-Maxwell law)
- $\Phi_E = EA_{\perp}$
- $\oint \vec{B} \cdot d\vec{l} = B(2\pi r) = \mu_0 \left[i + 0 \right]$
- $B = \frac{\mu_0 i}{2\pi r}$
- $\Phi_E = \frac{Q}{\epsilon_0} \pi R^2 = \frac{Q}{\epsilon_0}$
- $B(2\pi r) = \mu_0 \left[0 + \epsilon_0 \times \frac{1}{\epsilon_0} i \right]$
- $B = \frac{\mu_0 i}{2\pi r}$

Video player interface: 11:27 / 11:42

Calculating displacement & conduction current | Electromagnetic waves | Physics | Khan Academy

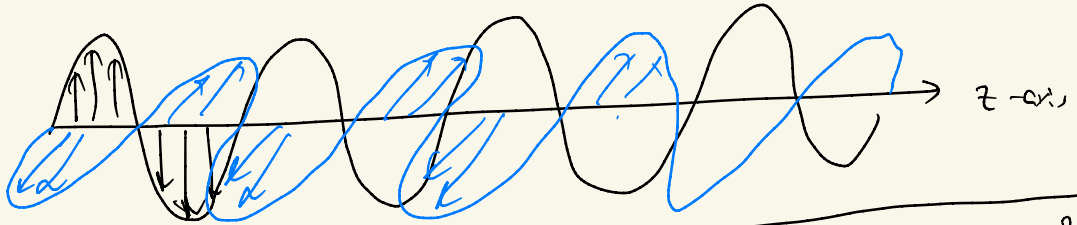
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Chapter 33: Electromagnetic Waves

Cber = EM Волна



$(\vec{E} \perp \vec{B}) \perp \vec{k}$
 ↑
 propagation vector

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 2.998 \times 10^8 \frac{\text{m}}{\text{s}}$$

$$v = \frac{1}{\sqrt{\mu \epsilon}} < c$$

Математически: wave number

$$k = \frac{2\pi}{\lambda}$$

